A-A-53916
26 June 1989
SUPERSEDING
MIL-B-37966
16 December 1985

COMMERCIAL ITEM DESCRIPTION

BUR, BONE (Wire Pass Drill)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-B-37966 which is cancelled.

Salient Characteristics:

This commercial item description covers wire pass drill type surgical burs of the following head diameters:

Size 1 - 1.0 mm (.0394 inches)
Size 2 - 1.5 mm (.0591 inches)
Size 3 - 2.0 mm (.0787 inches)

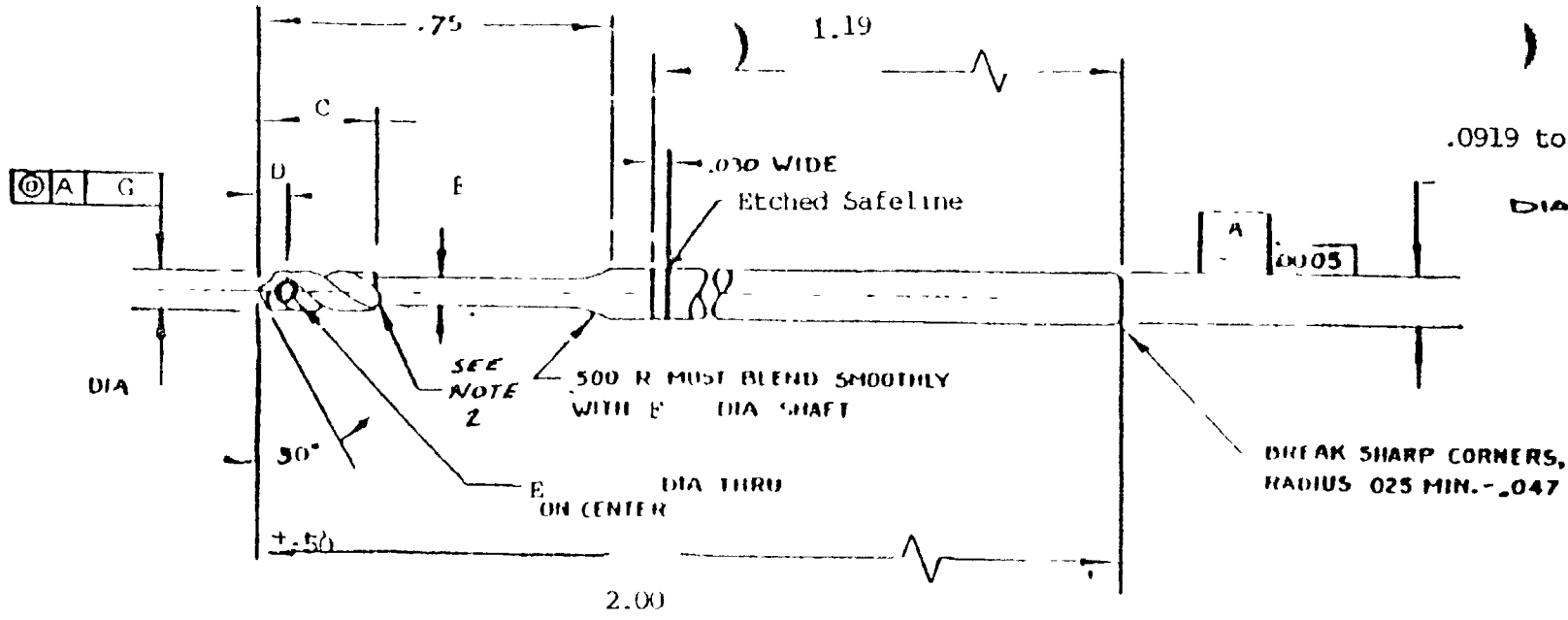
Material. The burs shall be fabricated from Class 5, Type 630, corrosion-resistant steel in accordance with ASTM F899.

Hardness. The burs shall be properly hardened and passivated. After proper stress relieving, the burs shall have a Rockwell hardness of C40 to C44, when tested as specified herein.

Style and dimensions. The burs shall be of one-piece construction and shall be furnished in the head diameter sizes specified in Figure 1. The burs shall be compatible for use with medium bur guard component of PNEUMATIC BONE DRILL (see notes). The proximal end of the burs shall be round and non-notched. The operative end cutting edges of the burs shall be uniformly cut, shaped and sharpened. The blades of the burs shall be sharpened to cut at the tip as well as the sides of the head and shall provide automatic chip clearance.

AMSC N/A FSC 6515
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SIZE	B	C	D	E	F	G
1	.040 +.002 -.000	.375 ±.060	.047 ±.010	.017 to .020	.040 +.002 -.000	.002
2	.060 ±.005	.250 ±.010	.063 ±.010	.030 ±.005	.051 ±.005	.001
3	.080 ±.005	.250 ±.010	.063 ±.010	.030 ±.005	.063 ±.005	.001

NOTES:

1. Head Shape: RH Cutting, End Cutting, 2 teeth, 20° RH Spiral teeth to cut across center
2. For Style 2. .12 Flute sweepout permissible on .051 diameter shank.
   For Style 3. .12 Flute sweepout permissible on .063 diameter shank.
3. Finish: #32
4. Remove all burrs.
5. Unless otherwise specified, dimensions are in inches.

FIGURE 1. Bur, bone

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Concentricity and straightness. The burs shall meet the concentricity and straightness requirements specified in Figure 1.

Runout. The total indicated runout of the burs shall not exceed 0.001 inch when measured as specified herein.

Marking. The burs shall be marked with a safeline as specified in Figure 1.

Finish. The finish of the burs shall be as specified in Figure 1. The operative end shall have a fine honed finish with sharp cutting edges. The safeline shall have an etched finish.

Corrosion resistance. The burs shall not show evidence of corrosion when tested as specified herein.

Workmanship. The burs shall have uniformly cut and formed teeth and shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One bur, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspection lot. An inspection lot, for purposes of sampling and testing in accordance with MIL-STD-105, shall consist of burs of the same size, manufactured under essentially the same conditions. Each unit package of burs shall be permanently marked with a serial number, or a combination of letters and numbers, which shall refer to the manufacturer's records for that particular lot or batch.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an insepection level of II. The unit of product for examination purposes shall be one bur.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The unit of product for test purposes shall be one bur. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. Sampling for dimensions tests shall be conducted at inspection level II, AQL 1.5 (percent defective). Sampling for runout and corrosion resistance tests shall be conducted at inspection level S-1, AQL 1.0 (percent defective). Sampling for hardness tests shall be conducted at inspection level S-1, and the acceptance number shall be zero for all sample sizes.

Sampling of raw materials. A sufficient quantity of each lot of raw steel used in the fabrication of the instruments shall be selected for chemical analysis.

Examination. The burs shall be examined to determine compliance with the requirements contained in this specification. Examination shall include, but shall not be limited to, the defects listed below.

Defects

Dimensions not as specified.
Design not as specified.
Bur not of one-piece construction.
Cutting edges not sharp.
Cutting edges not uniformly cut and shaped.
Finish of bur not as specified.
Instructions for use inaccurate, inadequate, or missing.
Cutting edges not properly angled.
Sizes not as specified.

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Defects (continued)

Laminations, cracks, bends, deformed portions, gouges, chips, or broken areas present in any part of bur.
Etched safeline missing, not clearly defined or not of uniform thickness.
Number of teeth not as specified for applicable bur size.

Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with the gages.

Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadowgraph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360 degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Microscopic examination. Examination for possible bur defects not perceptible to the unaided eye shall be made using a 10X magnification.

Tests.

Chemical analysis. In the absence of certificates of quality, the chemical analysis of the instruments shall be determined in accordance with method 111 or 112 of FED-STD-151. In the event of a conflict, method 111 shall govern.

Hardness. Hardness shall be determined in accordance with American Society for Testing and Materials (ASTM Designation E18 (Rockwell or E92 Vickers or Diamond Pyramid).

Corrosion resistance. The burs shall meet the requirements specified herein when subjected to the boil test as specified in ASTM 1089.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine of the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTES:

1. The following National Stock Numbers are covered by this document:

NATIONAL STOCK NUMBERS	ITEM IDENTIFICATION
6515-01-139-7741	BUR, BONE, Wire Pass Drill 1.5 mm Head Diameter
6515-01-139-7744	BUR, BONE, Wire Pass Drill 1.0 mm Head Diameter
6515-01-139-7745	BUR, BONE, Wire Pass Drill 2.0 mm Head Diameter

2. The following National Stock Number is referenced by this document.

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-01-116-8197	DRILL BONE, PNEUMATIC

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

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A-A-53916

MILITARY INTERESTS:

Custodians

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4632
Location- NCR Disk 1

06613

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(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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